Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 03)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
2	
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9	
10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 . 1 Explain the difference between approaches to sustainable lifestyles based on **moral belief** and on **enlightened self-interest**.

[2 marks]

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0 1 . 2 Explain why the overall environmental impact of humans depends on two main factors.

[2 marks]

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0 1 . 3 Give **one** activity that is sustainable with a small population but unsustainable if the population is too large.

[1 mark]

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0 2 A salmon farm rears fish in net cages in a sheltered sea loch. Over one production cycle the farm used 2 400 t of feed and produced 2 000 t of salmon.

0 2 . 1 Calculate the food conversion ratio (mass of feed per unit mass of fish produced) for the farm.

[2 marks]

Answer

Do not write outside the box

0 2 . 2 Explain why so much of the world's aquaculture is of carnivorous species such as salmon, and why this reduces the benefit to world food supplies.

[3 marks]

0 2 . 3 Explain **four** environmental impacts of rearing salmon in net cages in a sea loch.

[4 marks]

0 2 . 4 Explain **three** ways the environmental impacts of the farm could be reduced.

[3 marks]

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0 2 . 5 Explain why aquaculture cannot simply replace wild capture fisheries.

[3 marks]

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0 3 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 3 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 4 . 1 Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability .....

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Dissolved oxygen .....

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pH .....

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0 4 . 2 Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

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0 4 . 3 Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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0 5 A national conservation body must decide how to divide a limited budget between species.

0 5 . 1 Describe **three** criteria the IUCN uses when deciding the Red List category of a species.

[3 marks]

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0 5 . 2 A population of lemurs has fallen from 12 000 to 2 100 mature individuals. Calculate the percentage decline.

[2 marks]

Answer ..... %

0 5 . 3 Explain **three** reasons, other than its Red List category, why one species may be given a higher conservation priority than another.

[3 marks]

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0 5 . 4 Explain the difference between CITES Appendix I and Appendix II, and explain why controlling trade can protect a species more effectively than habitat protection alone.

[3 marks]

0 5 . 5 Describe **four** ways that UK legislation and designated areas protect habitats.

[4 marks]

15

0 6 . 1 Give **two** natural processes that are driven by solar energy.

[2 marks]

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0 6 . 2 Explain why natural processes can take place at low temperatures.

[2 marks]

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0 6 . 3 Give **one** human activity that uses high temperatures to achieve a result that natural systems achieve at low temperature.

[1 mark]

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0 7 Figure 1 shows a simplified Antarctic food web.

Figure 1

Trophic level	Organisms
Producers	Phytoplankton, ice algae
Primary consumers	Zooplankton, krill
Secondary consumers	Small fish, squid, baleen whales, crabeater seals, penguins
Higher consumers	Toothfish, albatross, leopard seal, killer whale

0 7 . 1 Explain why the sea around Antarctica is so productive despite the cold.

[3 marks]

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0 7 . 2 Use Figure 1 to explain why fishing for krill and for Patagonian toothfish threatens other species.

[3 marks]

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0 7 . 3 Explain the importance of Antarctica in controlling global natural cycles.

[3 marks]

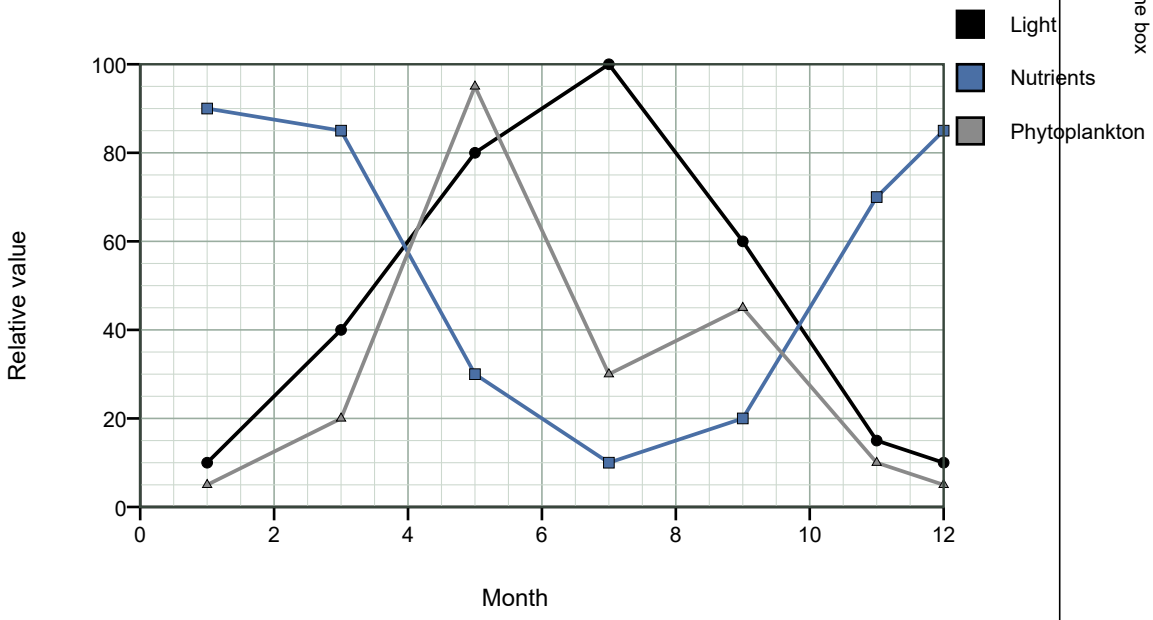
0 7 . 4 Explain why the area of winter sea ice around Antarctica has increased in recent years even though air and sea are warming.

[1 mark]

10

0 8 Figure 1 shows the seasonal changes in light intensity, nutrient concentration and phytoplankton biomass in the surface water of a temperate ocean in the northern hemisphere.

Figure 1



0 8 . 1 Use Figure 1 to explain the spring peak and summer decline in phytoplankton biomass.

[4 marks]

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0 8 . 2 Explain why tropical open oceans have low productivity all year despite high light and temperature.

[2 marks]

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0 8 . 3 Explain why areas of upwelling and coastal waters are the most productive parts of the sea.

[3 marks]

0 8 . 4 Explain why phosphate is often the limiting nutrient in the sea.

[1 mark]

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0 9 . 1 Explain what a carbon footprint should include when calculated for a car.

[3 marks]

0 9 . 2 Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

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0 9 . 3 Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

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1 0 . 1 Evaluate the relative importance of ex-situ and in-situ methods in the conservation of endangered species.

[9 marks]

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Extra space

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1 0 . 2 State what is meant by **rewilding**.

[1 mark]

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Write an essay on one of the following topics.

11 . 1 Discuss the methods that can be used to conserve biodiversity and evaluate their effectiveness.

[25 marks]

OR

11 . 2 Explain how the physical features of the Earth and the activities of living organisms have combined to create and maintain conditions suitable for life.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS